OFFLINE NETWORK RESILIENCE WORKSHOP CURRICULUM

A Three-Session Community Training Program

Facilitator Guide — All materials printable, no internet required.

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Introduction for Facilitators

This curriculum is designed to be delivered in three standalone sessions of approximately 2.5–3 hours each. Sessions can be run on consecutive weekends or spread over a month. Each session is self-contained enough that attendees who miss one can still benefit from others.

Materials Needed (All Sessions)

- This curriculum guide (facilitator copy, printed)
- Participant handout packet (print one per person — Sections marked **PARTICIPANT**)
- Quick-cards set (pre-printed and laminated — 1 set per 2 participants)
- Whiteboard or flip chart + markers
- Extension cords and power strips
- Optional: projector or large monitor

Session Equipment Checklist

Verify all items below before each session begins. Mark each item as confirmed.

☐	Item	Notes
☐	Baofeng radios or similar	Minimum 4 units
☐	Meshtastic devices	Minimum 2 — ideally 4+
☐	Laptop or tablet with Meshtastic app	Pre-installed preferred
☐	USB kit drives	1 per participant or 1 per group
☐	Printed field manual	At least 2 copies
☐	Printed quick-card sets (laminated)	1 set per 2 participants
☐	Solar panel + battery demo kit	Optional but recommended
☐	Go-Box demonstration unit	Optional

SESSION 1

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FOUNDATIONS AND RADIO BASICS

Duration:

2.5

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3 hours

Maximum Participants:

20 (larger groups work but reduce hands-on time)

Prerequisites:

None

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this is the entry point

▶ SESSION 1 LEARNING OBJECTIVES

By the end of this session, participants will be able to:

1. Explain why offline communication networks matter and when they're needed
2. Identify the three pillars of community communications resilience
3. Operate a basic VHF/UHF handheld radio (power on, frequency, transmit, receive)
4. Perform a proper radio check-in using standard procedure
5. Locate and use the Emergency Frequency Quick-Card

SEGMENT 1.1

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WELCOME AND INTRODUCTIONS

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20 Minutes

Facilitator Actions

- Welcome participants; introduce yourself and any co-facilitators
- Have participants introduce themselves (name, neighborhood, any radio experience)
- Display or draw the three-pillar diagram: **Mesh** / **Radio** / **Physical**

★ ICE BREAKER ACTIVITY — "What Would You Miss Most?" (5 minutes)

Ask: *"If the internet and cell phones went down right now, what communication would you miss most — and how would you replace it?"*

Brief discussion (5 minutes). Record answers on whiteboard.

Key Teaching Point: Most of what we "need" from the internet can be replicated locally with the right preparation. This session is about how.

✎ PARTICIPANT HANDOUT 1.1 — THE CASE FOR RESILIENCE

Three Scenarios Where This Matters

Scenario

Description

Core Challenge

A — The Ice Storm

A major ice storm knocks out power for 10 days. Cell towers run on generator for 6 hours, then go dark.

How do you find out who needs help? How do you coordinate resources?

B — The Earthquake

A 6.5 magnitude earthquake disrupts fiber optic lines and overwhelms cell networks

with traffic. Official emergency broadcasts are sporadic.

Neighbors two blocks away might as well be in another city.

C — The Cyberattack

A ransomware attack on regional ISPs knocks out internet for a major metropolitan area for 72 hours. 911 dispatch is impacted.

How does your neighborhood self-organize?

Discussion Questions

1. Which scenario feels most realistic for your community?
2. What resources do you already have that could help?
3. Who in your neighborhood would you turn to first?

SEGMENT 1.2

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UNDERSTANDING THE THREE PILLARS

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25 Minutes

✎ **FACILITATOR NOTE**

Draw or display the three pillars as a visual throughout this segment. Keep the diagram visible on the whiteboard for the entire session.

Pillar 1 — Mesh Networks (8 minutes)

- **Definition:** Devices talk directly to each other — no internet required

- **Self-healing:** If one node goes down, traffic routes around it
- **Range:** LoRa mesh can reach 1–15 km depending on terrain
- Text-based messaging, GPS position sharing, limited data

▶ DEMONSTRATION

If Meshtastic devices are available, turn on two and show a message being sent without any internet or cell connection. Have a participant send a message.

Pillar 2 — Radio Communications (10 minutes)

- Radio has worked for 100 years without internet
- Multiple layers: handheld (local), mobile (neighborhood), HF (regional/national)
- Licensed vs. unlicensed bands — brief overview
- Why amateurs are the backbone of emergency comms

▶ DEMONSTRATION

Power on a radio. Scan for local repeaters. Demonstrate a radio check.

Pillar 3 — Physical Distribution (7 minutes)

- When everything electronic fails, humans are the network
- Runners, bulletin boards, printed materials
- Pre-agreed meeting places and times
- The oldest and most reliable system

▶ GROUP ACTIVITY

Ask the group: *“Where would you put a community bulletin board in your neighborhood that would reach the most people? Could everyone walk to it?”*

SEGMENT 1.3

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HANDS-ON RADIO BASICS

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45 Minutes

Materials: 4+ handheld radios (Baofeng UV-5R or similar)

Part A — Radio Orientation (15 minutes)

Walk through with all radios powered on:

1. Power button and volume control
2. Channel/frequency display
3. PTT (Push-to-Talk) button
4. Monitor/scan function
5. Squelch — what it is and how to set it
6. Menu navigation — basic settings

✎ PARTICIPANT HANDOUT 1.3A — RADIO ANATOMY LABELS

[Simple labeled diagram of a handheld radio with arrows pointing to each control. Print one per participant and have them label the parts as the facilitator walks through each one.]

Part B — Standard Radio Procedure (15 minutes)

Standard Format

Step	Action	Example / Notes
1	Listen before transmitting	Wait at least 3 seconds to confirm channel is clear
2	Identify yourself and destination	“[Your call sign/name], [destination call sign/name]”
3	Keep transmissions short	Say what you need; avoid rambling
4	Say “over”	When waiting for response
5	Say “out”	When finished with the exchange
6	Say “break break break”	For emergencies — interrupts all traffic immediately

Role Play Exchanges (Pairs Practice)

Exchange	Type	Scenario
1	Routine	Request for information about road conditions
2	Urgent	Reporting a neighbor who needs a welfare check
3	Emergency	Reporting a structure fire with people inside

Part C — Phonetic Alphabet Drill (15 minutes)

Distribute Quick-Card 1. Practice saying names using phonetics:

- Facilitator says a word; participants spell it using phonetics
- Then do locations: street names, neighborhood names
- Finally: practice call signs

▶ ACTIVITY — Speed Round

Random letters called out; participants respond with phonetic words as fast as possible. Increase pace each round to build reflex recall.

> SEGMENT 1.4
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> LICENSE OVERVIEW AND NEXT STEPS
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> 20 Minutes

✎ PARTICIPANT HANDOUT 1.4 — LICENSE ROADMAP

License

Cost

Time to Get

What You Can Do

Who It's For

FRS (none required)

$0

Immediate

Local comms, limited power

Everyone

GMRS

$35 / 10 yr

1–2 weeks

Better local comms, family use, repeater access

Households

Technician

~$15 exam

2–4 weeks study

Full VHF/UHF, all emergency frequencies

Serious operators

General

~$15 exam

Additional study

HF regional/national communications

Net coordinators

✎ FACILITATOR NOTE

Emphasize that FRS radios are valuable right now even without any license. Getting a Technician license expands options significantly and the exam is achievable with 2–3 weeks of study. Research and insert local exam session dates before delivering this session. Offline study resources are available on the USB kit.

SEGMENT 1.5

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SESSION WRAP-UP AND HOMEWORK

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10 Minutes

Recap and Distribute

- Recap the three pillars: Mesh, Radio, Physical
- Distribute: Quick-Card set (Cards 1 and 2 particularly)

Homework Assignments (Optional but Encouraged)

7. Program the local repeater frequency into your radio (use Quick-Card 2)
8. Find 3 neighbors who might be interested in this training
9. Begin studying for Technician license (HamStudy.org or apps — offline resources on USB)
10. Identify a location in your neighborhood for a community bulletin board

▶ PREVIEW — SESSION 2

Mesh networks hands-on, USB kit setup, data resilience. Participants who attend Session 2 will configure real devices.

✎ SESSION 1 EVALUATION FORM — Collect before participants leave

Question

Response

What was most valuable?

What questions do you still have?

Would you like to be contacted about Session 2?

☐ Yes ☐ No

Contact information

SESSION 2

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MESH NETWORKS AND DATA RESILIENCE

Duration:

2.5

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3 hours

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Maximum Participants:

20

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Prerequisites:

Session 1 (recommended but not required)

▶ SESSION 2 LEARNING OBJECTIVES

By the end of this session, participants will be able to:

1. Explain how a mesh network functions at a practical level
2. Set up and configure a Meshtastic device
3. Navigate the USB kit folder structure and access offline resources
4. Identify a suitable location for a community mesh node
5. Describe the data security practices for offline systems

SEGMENT 2.1

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REVIEW AND CONTEXT

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15 Minutes

- Quick review of Session 1 key points (5 minutes)
- New participant orientation (5 minutes — for those who missed Session 1)
- Preview of today's hands-on activities (5 minutes)

▶ OPENING DISCUSSION

"What information would you most need access to in a 10-day outage?" Record

answers on whiteboard. Refer back throughout session to anchor lessons in real need.

> SEGMENT 2.2
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> MESH NETWORKING DEEP DIVE
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> —
>
> 30 Minutes

Part A — How Meshtastic Works (15 minutes)

Concept	Plain-Language Explanation
LoRa Technology	Long Range radio — like Wi-Fi but much longer range, low data rate
Flood Routing	Each device receives and retransmits messages to extend coverage
Store-and-Forward	Messages held at a node until they can be passed along
No Infrastructure	No internet, no servers, no accounts needed — fully standalone
GPS Position Sharing	Built in and optional; share or hide your location
Message Size	Text messaging only (no voice) — up to ~228 bytes per message

▶ VISUAL DEMONSTRATION

Use 2–4 Meshtastic devices placed around the room. Show messages routing through intermediate nodes. Point out the node list on the app to illustrate the

mesh forming in real time.

Part B — Network Planning Exercise (15 minutes)

▶ GROUP ACTIVITY — “Map Your Neighborhood Mesh”

Materials: Large printed neighborhood map (or hand-drawn), sticky notes, markers

Instructions (groups of 4–5):

1. Groups of 4–5 people take a map section
2. Mark high points (hills, tall buildings, church steeples, water towers)
3. Mark potential node locations (someone who could host a node)
4. Identify coverage gaps
5. Note where the most vulnerable community members live

Discussion: What's the minimum number of nodes to cover your neighborhood? Where are the critical relay points?

SEGMENT 2.3

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HANDS-ON MESHTASTIC

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45 Minutes

Materials: 4+ Meshtastic devices, smartphones with Meshtastic app

Part A — Device Setup (20 minutes)

Walk through step by step (all participants follow along):

11. Install Meshtastic app (or use pre-installed version from USB)
12. Power on device — observe LED indicators
13. Pair via Bluetooth
14. Set region: **United States**
15. Set device name (use neighborhood + number, e.g., “Riverside-07”)
16. Confirm channel: LongFast (default) or community-preset channel
17. Send a test message to the group

✎ PARTICIPANT HANDOUT 2.3A — MESHTASTIC SETUP CHECKLIST

☐	Step	Confirmed By
☐	App installed and open	
☐	Bluetooth enabled on phone	
☐	Device powered on (LED indicator active)	
☐	Paired to device (appears in app)	
☐	Region set to United States	

☐

Device name set

☐

Channel confirmed (LongFast or community channel)

☐

Test message sent and received

☐

GPS enabled (optional — confirm if desired)

Part B — Range Test Exercise (25 minutes)

Pair up participants. Give each pair a Meshtastic device. Send pairs to different locations:

- Inside the building (various rooms)
- Outside (various distances)
- One pair stays as “base station”

Each pair sends a message every 5 minutes with their location and signal quality. Record results on whiteboard map.

✎ FACILITATOR NOTE — Debrief Prompt

What affected range? Lead discussion around: wall composition, distance, elevation changes, physical obstacles. Reinforce that node placement on rooftops or upper floors dramatically improves coverage.

> SEGMENT 2.4

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> THE USB KIT
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> 30 Minutes

Part A — USB Kit Tour (15 minutes)

Distribute USB drives (one per participant or group). Walk through the folder structure together:

Folder	Contents	How to Use
📁 **MANUALS/**	Field manual PDF	Open with any PDF viewer; navigate offline
📁 **MAPS/**	Offline map files	Open in OsmAnd or Maps.me (no internet needed)
📁 **SOFTWARE/**	Meshtastic flasher, Briar, Kiwix	Install from USB; no download required
📁 **MEDIA/**	Training video clips, quick-card PDFs	Play locally; print quick-cards from PDF
📁 **DATA/**	Frequency list, contact directory format	Reference during operations; keep updated
📁 **TEMPLATES/**	Message forms, log sheets	Print as needed; fill by hand in field

⚠ **KEY TEACHING POINTS — USB Kit**

- These USB drives work with **no internet**
- Keep one at home; keep one in your go-bag
- Update on the **first Sunday of each month** at community update sessions
- The Kiwix app + Wikipedia ZIM = offline encyclopedia for medical, technical, and legal reference

Part B — Offline Resources Practice (15 minutes)

Participants practice independently:

18. **Task 1:** Navigate to your neighborhood on the offline map
19. **Task 2:** Find the field manual and navigate to the frequency reference
20. **Task 3:** Open Kiwix and search for “first aid” (or similar practical topic)
21. **Task 4:** Locate and print-preview the information runner message form

> SEGMENT 2.5
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> DATA SECURITY BASICS
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> 15 Minutes

✎ PARTICIPANT HANDOUT 2.5 — DATA SECURITY QUICK GUIDE

What to Store

How to Store

What to Avoid

Public docs (manual, maps, frequencies)

Plain files on USB

Names and addresses stored together

Community contacts

7-Zip AES-256 encrypted archive

Passwords, financial information

Mesh channel key

Share verbally or on paper — in person only

Posting online or in plain text files

Additional topics to cover in this segment:

- What **NOT** to put on community USB drives: passwords, financial info, medical records, personal identifiers
- How to create an encrypted archive using 7-Zip AES-256 — live demonstration
- Meshtastic channel encryption — AES-256 pre-shared key setup
- Physical security: do not leave USB drives unattended at public events

> SEGMENT 2.6
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> SESSION WRAP-UP
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> 15 Minutes

- Recap key skills: mesh setup, USB navigation, data security
- Distribute: Quick-Cards 3, 7, 9 (Meshtastic, Node Check, Offline Data)

Homework

22. Install Meshtastic on a device you own
23. Identify 2 neighbors with elevated locations for potential nodes

24. Make a copy of your USB kit for a household member

▶ PREVIEW — SESSION 3

Community organization, drills, go-box build, and a simulated blackout exercise. Participants will take on real roles in a tabletop scenario.

SESSION 3

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COMMUNITY ORGANIZATION AND RESILIENCE EXERCISES

Duration:

3 hours

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Maximum Participants:

25

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Prerequisites:

Sessions 1 and 2 (recommended)

▶ SESSION 3 LEARNING OBJECTIVES

By the end of this session, participants will be able to:

1. Assign and understand community resilience roles
2. Design a basic communication tree for their neighborhood cell
3. Conduct a basic after-action review
4. Operate a deployed go-box setup
5. Facilitate a monthly radio net

SEGMENT 3.1

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REVIEW AND ASSESSMENT

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20 Minutes

Brief quiz (ungraded, for self-assessment): 10 questions covering Sessions 1 and 2 material. Review answers as a group — clarify any confusion before proceeding.

★ CELEBRATION

Recognize anyone who has gotten licensed since Session 1! Acknowledge effort publicly — this encourages others to pursue licensing.

SEGMENT 3.2

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COMMUNITY ORGANIZATION STRUCTURE

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30 Minutes

Part A — Roles and Responsibility Assignment (15 minutes)

Present the six core roles (from Section 7 of the field manual):

#	Role	Primary Responsibility	Assigned To
1	**Network Coordinator**	Overall coordination of the community network; primary contact for external agencies	
2	**Node Custodians**	Maintain mesh nodes; ensure power and connectivity; report outages	
3	**Radio Operators**	Licensed operators who manage voice radio traffic and nets	
4	**Information Runners**	Physical message delivery between nodes when electronic comms are unavailable	
5	**Supply Custodian**	Track and maintain shared equipment inventory; manage go-box	
6	**Training Coordinator**	Schedule sessions, onboard new participants, coordinate USB updates	

▶ GROUP ACTIVITY — Role Matching

Each participant receives a card describing a scenario/skill set. They self-select or are placed into roles based on skills and interest. Not everyone needs a formal role — many people are simply “network participants.” Fill in the community roles table together on the whiteboard. Capture in the participant directory template.

Part B — The Neighborhood Cell Structure (15 minutes)

Explain cells: 10–20 households, one block captain, direct communication with neighbors.

▶ ACTIVITY — “Draw Your Cell”

Participants sketch their immediate neighborhood (their block/street/cul-de-sac). Mark:

- Their own home
- Households they already know
- Households with elderly/disabled residents (priority for check-in)
- Potential bulletin board locations
- Best route to the community hub

SEGMENT 3.3

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COMMUNICATION TREE BUILD

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25 Minutes

Explain the tree structure: each person calls/radios 3–5 others; they each call 3–5 more. A message can reach 100 people in 2–3 “hops.”

▶ GROUP ACTIVITY — “Build the Tree”

Using sticky notes on a wall or whiteboard:

1. Place facilitator at top
2. Each participant places their note connected to who would contact them
3. Fill gaps — who has no one to call them?
4. Identify single points of failure (someone who is the only link to 10 people)
5. Add redundant connections

Discussion: What happens when someone in the tree is unavailable? Plan for substitutes before it matters.

Take a photo of the completed tree. Transcribe to the template in TEMPLATES/ folder on USB.

SEGMENT 3.4

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GO-BOX DEMONSTRATION AND HANDS-ON

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30 Minutes

Materials: Go-box (pre-assembled or components laid out)

Go-Box Component Checklist

☐	Component	Notes
☐	VHF/UHF radio (programmed)	Pre-program all local frequencies before deployment
☐	Antenna + coax	Verify connector compatibility
☐	Battery (12V LiFePO4 recommended)	Check charge before each deployment
☐	Solar panel + charge controller + cables	Test outdoors before relying on in emergency
☐	Meshtastic device	Pre-configured with community channel
☐	Laptop/tablet	USB kit loaded; Meshtastic app installed
☐	USB kit	Current version; updated within 30 days
☐	Quick-card set (laminated)	Full set in protective sleeve
☐	Logbook + pens	At least 2 pens; waterproof preferred
☐	License copy	Laminated copy of operator license(s)
☐	First aid kit (basic)	Check expiry dates annually

Deployment Sequence (Quick-Card 12)

25. Select location (elevation, line-of-sight, access)
26. Deploy antenna
27. Connect power
28. Power on radio
29. Monitor — confirm channel activity
30. Check in using standard procedure

▶ HANDS-ON ROTATION

Participants rotate through: setting up the antenna, connecting the battery, and performing a radio check using proper procedure. Each rotation approximately

8–10 minutes.

SEGMENT 3.5

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SIMULATED EXERCISE

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35 Minutes

▶ “BLACKOUT SCENARIO” — Tabletop + Partial Activation

Scenario: *It is Day 2 of a major power outage. Cell networks are down. You have only radio and mesh communications available. Reports are coming in that an elderly resident on Maple Street has not been seen since before the storm.*

Participants play their assigned roles. Facilitator injects events and observes without intervening.

Inject	Time	Event	Expected Response
INJECT 1	Minute 5	Report of elderly resident not responding to door knocks on Maple Street	Welfare check assigned; runner dispatched or radio contact made
INJECT 2	Minute 15	Runner arrives with written note from across the neighborhood — food shortage at the community center	Information logged; resource coordinator notified; runner acknowledged
INJECT 3	Minute 25	Radio report of possible gas leak	Emergency break used; area alerted;

Inject	Time	Event	Expected Response
		two blocks away	external contact attempted via net

After-Action Review (AAR) — Quick-Card 11 Format

AAR Question	Group Notes
What was planned?	
What happened?	
What went well?	
What needs improvement?	
Action items?	

> SEGMENT 3.6
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> MONTHLY NET PRACTICE
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> 20 Minutes

Simulate a community radio net. Designate one participant as Net Control.

✎ NET CONTROL SCRIPT

“[Time] hours local. This is the [Community Name] Emergency Communications Net. Net Control is [name/call sign]. All stations please check in with your call sign, location, and status. [First station], go ahead.”

- Check-in procedure for all participants
- Pass one piece of “traffic” (a simulated message)
- Close the net formally

> **✎ FACILITATOR NOTE — Debrief**
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> What was awkward? What needs practice? Use this debrief to schedule a real monthly net. Identify a willing Net Control volunteer before participants leave.

> SEGMENT 3.7
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> NEXT STEPS AND PROGRAM CLOSE
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>
> 20 Minutes

30-60-90 Day Action Plan

30 DAYS

☐	**Action Item**
☐	All participants with radios have joined one real radio net
☐	Node custodians identified for at least 2 community nodes
☐	Communication tree built and tested
☐	USB kits distributed to all participants

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60 DAYS

☐	Action Item
☐	First community mesh nodes deployed and operational
☐	Monthly net established and running
☐	Block captains identified for neighborhood cells
☐	Go-box assembled and available

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90 DAYS

☐	Action Item
☐	Full network coverage across target area
☐	First full tabletop exercise conducted
☐	Printed materials (manual, quick-cards) distributed to all households
☐	Update process established (monthly USB update sessions)

Community Commitment Card

✎ PARTICIPANT COMMITMENT CARD — Sign before leaving

I commit to the following as a member of this community resilience network:

☐

Commitment

☐

Share this training with at least 3 neighbors

☐

Maintain my assigned role/equipment

☐

Participate in monthly net checks

☐

Attend annual full exercise

Signed: _____ Date: _____

Facilitator Resources

Common Questions and Answers

Q: Do I need a license to use a mesh network?

No. Meshtastic operates in the unlicensed 915 MHz ISM band. No FCC license is required.

Q: What if someone in my neighborhood refuses to participate?

You cannot compel participation. Focus on those who are willing. Having even 20% of households connected provides significant benefit for the whole community.

Q: Is this legal?

Yes. All technologies covered in this curriculum are legal when used as described. Amateur radio requires a license for licensed bands. FRS/GMRS are legal with appropriate licensing. Mesh networks in the ISM band require no license.

Q: What about privacy?

Meshtastic messages on the default channel are not encrypted. Use the community channel key for sensitive communications. USB drives should not contain sensitive personal information in unencrypted form.

Q: How do we fund this?

Many community resilience networks operate entirely on volunteer time and donated equipment. Some obtain grants through local emergency management

offices, community foundations, or mutual aid networks. Equipment costs can be shared — one community node can serve 20+ households.

Supplemental Workshop — Advanced Topics

> **▶ OPTIONAL 2-HOUR ADD-ON SESSION**
>
> **Prerequisite:** Technician or General license for HF/Winlink topics.

Topic	Duration	Prerequisite
HF Radio and Long-Distance Communications	30 min	General license recommended
Winlink and JS8Call for digital messaging over radio	30 min	Technician or General license
APRS for position tracking and messaging	20 min	Technician license
Raspberry Pi mesh node build	40 min	None (technical interest helpful)

Offline Network Resilience System — Workshop Curriculum

This curriculum is part of the Offline Network Resilience System. **Reproduce and adapt freely for your community.**

Facilitator questions: Contact your local ARES/CERT coordinator or amateur radio club.

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